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Author: Dário Hipólito Margarida Santos-Reis Luís M. Rosalino

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Effects of agro-forestry activities, cattle-raising practices and food-related factors in badger sett location and use in Portugal

Dário Hipólito, Margarida Santos-Reis & Luís M. Rosalino*

CE3C - Centre for Ecology, Evolution and Environmental Changes, Faculdade de Ciências, Universidade de Lisboa, 1749-016 Lisboa, Portugal (DH: dariohipolito@gmail.com; MSR: mmreis@fc.ul.pt; LMR: lmrosalino@ua.pt)

*current affiliation: CESAM & Department of Biology, University of Aveiro, Campus Universitário de Santiago, 3810-193 Aveiro, Portugal

Corresponding author: Luís Miguel Rosalino, CESAM & Department of Biology, University of Aveiro, Campus Universitário de Santiago, 3810-193 Aveiro, Portugal (e-mail: lmrosalino@ua.pt; Telephone: +351 234 370 350)

Abstract

Mediterranean landscapes in Europe are characterised by a mixed matrix of agriculture, agro-forestry or cattle-farming areas, which have influenced native communities for centuries. Recently, new changes were imposed on these agro-forestry landscapes due to novel management options that provide new challenges for wildlife. From a conservation perspective, there is an urgent need to assess what are the impacts and ecological responses of local wildlife populations to those changes. In the present study, we assessed the influence of human-caused disturbances (e.g. agroforestry practices, cattle-breeding activities and game management), together with food/water and landscape-related factors, on sett site selection by European badgers. We also tested the role of these factors, together with climatic variability, on sett use. We detected that areas further away from water sources, but close to game species feeding structures, have a higher probability of hosting a European badger sett. Moreover, sett use was promoted by low disturbance: setts with lower cattle presence and without understory removal activities were more often in use. Thus, our study shows that, in our study area, agro-forestry and cattle breeding activities constrain badger sett use, while food and water availability influence their distribution. Although badgers have been adapting to anthropogenic activities for centuries, they respond differently depending on the type and intensity of management. Thus, to ensure biodiversity persistence and sustainable human exploitation of the landscape, we need to identify trade-offs between wildlife ecological adaptations and anthropogenic agriculture, forestry and cattle production.

Keywords: *Meles meles*; Multi-use forests; Sett selection; Forest management; Southwestern Europe.

Introduction

Cork (*Quercus suber*) and/or holm (*Q. ilex*, *Q. rotundifolia*) woodlands are savannah-type agro-forestry-pastoral systems, termed "montado" in Portugal or "dehesas" in Spain (Bugalho et al., 2011). Agricultural, forestry and cattle-breeding practices and management changes, together with the recent introduction of other anthropogenically-managed activities, such as commercial and recreational hunting (Casanova and Memoli, 2001), have impacts on the diverse wildlife community that inhabits these forest systems (Bugalho et al., 2011). These impacts may lead to alterations in population patterns and processes (e.g. decreasing abundance – Pita et al., 2009; local extinctions - Cabezas-Díaz et al., 2009; dependence on anthropogenic resources - Rosalino et al., 2005a). The degree of impact depends on the magnitude of the driver of change and also on species resilience. Species that are capable of persisting in these human-shaped forest ecosystems are already resilient to landscape humanisation, but recent alterations impose new challenges to which they must respond. Among species that have managed to survive in agriculture landscapes, carnivores are particularly important since they are key players in ecosystem structure and functioning due to their ecological role (e.g. population regulators, pest controllers, seed dispersers; Correia, 2001; Herrera, 1989; Jaksić and Delibes, 1987) and their sensitivity to human activities (e.g. direct persecution, hunting; Muñoz-Igualada et al., 2008). Particularly important are those with large-scale patterns of distribution and abundance, such as red fox (*Vulpes vulpes*), stone-marten (*Martes foina*) or the European badger (*Meles meles*).

The European badger is one of the most common mesocarnivores living in southwestern Mediterranean cork/holm oak woodlands (Rosalino et al., 2005c). The ecological pattern and role of this carnivore is anchored in its social structure, with group size and composition varying according to available resources (Kowalczyk et al., 2003; Revilla and Palomares, 2002). In western Mediterranean environments, badgers live in small groups (3-4 adults and 3-4 cubs; Rosalino et al., 2004) and in low abundance; these being regulated by the availability of sites to build their communal dens, commonly known as setts (Rosalino et al., 2005b). Unlike most mustelid species and other carnivores, badgers build their own setts, which often comprise several entrances connected by a complex tunnel system (Roper, 1992), and their location is strongly influenced by physiographic factors (e.g. altimetry, geology; Rosalino et al., 2005b), proximity to feeding grounds (Good et al., 2001), habitat shelter characteristics (Virgós and Casanovas, 1999) or intensity and type of anthropic disturbance (e.g. human settlements, roads - Jepsen et al., 2005). In addition, sett use also varies greatly with the type of sett (main vs secondary - Loureiro et al., 2007), sett ectoparasitic load (Butler and Roper, 1996) or human disturbance (e.g. poaching activities, degree of public access - Jenkinson and Wheater, 1998).

Despite the species' sensitivity to disturbance, in Mediterranean areas badgers occupy areas where the landscape matrix consists of agricultural fields and/or agro-forestry systems but, to our knowledge, the influence of land management practices on badger sett location and use is seldom evaluated (but see Virgós and Casanovas, 1999), in spite of the large number of studies targeting several ecological facets of its southern populations (see Roper, 2010 for a review).

The objective of this study is to test if factors associated with human activities, such as agriculture, forestry, cattle breeding and game-feeding, influence badger sett location (sett vs non-sett sites) and levels of use (monthly use vs non-use), in an intensively managed agro-forestry farmstead. We tested four hypotheses, namely that badger sett locations will be determined by factors associated with: 1) food and water availability; 2) land cover; 3) disturbance; 4) or a combination of any of those. We hypothesize that badger setts will have a higher probability of being located in areas with less anthropogenic disturbance (e.g. low tree and cork harvesting, low understory clear-cutting, greater distance to rural houses, etc.), and greater access to basic resources (e.g. food, water and dense vegetation for refuge), when compared to non-sett sites. We also tested five hypotheses regarding factors determining sett use: 1) water availability; 2) land cover; 3) disturbance; 4) climate; 5) or a combination of any of those. We hypothesize that sett use throughout the year will decrease with increasing direct disturbance (e.g. understory clear-cutting, cattle presence) and changes in climatic conditions (e.g. higher precipitation, lower temperatures).

Material and Methods

Study area

The study was conducted in central west Portugal, at the "Charneca do Infantado" (Fig. 1), part of "Companhia das Lezírias, S.A." (hereafter CL), the largest Portuguese agro-forestry farmstead, located ca. 30km from Lisbon in Samora Correia and Alcochete counties. Charneca do Infantado comprises 10,000 hectares of agro-forest farmland. The area is characterized by poor, sandy and sandy-loam soils with low permeability and a typical Mediterranean climate, with dry and hot summers (temperatures reaching 40 ^{o}C) and cold and rainy winters (up to 400 mm of rain).

Cork oak woodlands, with varying shrub densities, are the main land use type (6,725 ha). Other forested areas include plantations of Maritime pine (*Pinus pinaster*; 971 ha), stone pine (*Pinus pinea*; 508 ha) and eucalyptus (*Eucalyptus globulus*; 476 ha). Agricultural activities are focused mainly on rice (630 ha), animal fodder (in irrigated areas - 460 ha), and olive (59 ha) and vine production (140 ha).

Another important economic activity in Charneca do Infantado is cattle production. This occurs on natural or permanent biodiverse pastures, many within and below cork oak trees, but patches of mixed forest with a dense shrub understory have been fenced for cattle exclusion at different time periods to improve tree regeneration and biodiversity preservation. The cattle are organized into herds that rotate among grazing plots during the wet season (from September/October until February/March depending on precipitation values). In the dry season herds are moved to “Lezíria”—another sector within CL—where pastures and rice are the main land cover types (2200 ha).

Besides agriculture and cattle breeding, one of the company’s main activities is forestry; focused on cork extraction and tree pruning in cork oak woodlands, wood extraction in pine stands, as well as stone pine cone and nut harvesting. Seasonal (September to May) hunting tours (e.g. for wild boar *Sus scrofa*, fox *Vulpes vulpes*, rabbit *Oryctolagus cuniculus*, partridge *Alectoris rufa*, and other birds such as pigeons *Columba* spp. or European turtle doves *Streptopelia turtur*) complement the economic activities within Charneca do Infantado. As a game management action, one hundred game-feeding stations were established for supplementary feeding (e.g. wheat provision), and these also benefit other wild species, such as badgers (pers. obs.).

Field methods

Badger sett surveys were carried out from May 2013 to May 2014. Based on a detailed land-cover map (Gonçalves et al., 2011), the area was divided into patches of more or less uniform land cover (N=34) and setts were sought by establishing linear transects with length proportional to the size of the patch and walked by 4-5 researchers spaced ca. 10m apart. All of the most extensive patches were surveyed, representing

30% of the total area (i.e. 3000ha¸ 24 patches). The remaining area was comprised of less adequate land cover types (e.g. rice fields, vineyards, reservoirs) or small-sized patches, which were not surveyed due to time constraints.

Badger setts were identified on the basis of signs of species' presence, such as latrines with scat, footprints or fur (Roper 2010). Each sett was characterized using several variables grouped into three main categories: land cover, soil disturbance and food/water availability (Table 1; see Supplementary Material 1 for full descriptions of all variables). Variable selection was based on previous studies on species behaviour and ecological requirements (e.g. Rosalino et al., 2005b; Virgós and Casanovas, 1999), and on characteristics potentially influencing sett location at the study area (e.g. clear-cutting activities and cattle-grazing areas). Entrance holes separated by less than 100m were considered as belonging to the same sett and were most probably connected underground, and therefore we excluded them from analyses (i.e. we excluded the structure with the lowest number of entrances). We used two scales for sett characterisation: 1) a local scale representing the characteristics of the specific site where the sett was located and defined by the Minimum Convex Polygon encompassing all entrances (i.e. the habitat type where the sett was located; Table 1); and 2) a core-area scale represented by a buffer of 350 m radius around the sett (i.e. the most abundant habitat within the core area). This buffer radius represents that of a circular mean core area, derived from the average core area size previously described for this species in southwestern Portugal (0.38 km^2; Rosalino et al., 2004).

Simultaneously, we randomly selected non-sett locations (ratio of 60 random points per 40 setts) in the surveyed area using a GIS environment (Quantum GIS, version 2.4.0-Chugiak; QGIS Development Team, 2014) and three exclusion criteria: i) < 350m from the farmstead limits (to avoid characterising areas outside the defined study area); ii) < 700m from known setts (to minimise spatial autocorrelation of circular buffers around each random point or sett); iii) flooded areas and dirt roads as candidates for random points. After selection, each random point was surveyed to ensure that no sett was located within the buffer limits.

When located, setts were monitored monthly (i.e. visited once a month) to assess use patterns (used *vs* not used), on the basis of recent signs of presence (e.g. fresh scat or footprints). Additionally, several variables were registered each month, namely: i) number of individual cattle in the grazing patches where identified setts were located, ii) any evidence of tree/shrub clear-cutting or hunting activity in the 15 days previous to the visit, and iii) climatic data (e.g. precipitation, mean monthly temperature) from the farmstead weather station (Table 2; see Supplementary Material 2 for full descriptions of variables).

Data analysis

Sett location

All variables that characterized the environment of setts at the local and core area scales (i.e. 350 m radius), where used simultaneously in this analysis because sett location may be influenced by factors acting at both scales or by their interaction. To avoid the bias associated with spatial clustering of setts and random points, we tested for spatial autocorrelation by using the Moran I index (Diniz et al., 2003), available in the R (R Core Team, 2014) package “ape” (Paradis et al., 2004). We also calculated the nearest neighbour index (NNI), or R (Clark and Evans, 1954), using Quantum GIS tools to assess if setts were clustered, or randomly or regularly distributed. The significance of the distribution was tested using a z-score test (Zar, 2010). Since we considered the influence of several categorical variables associated with disturbance and land cover (see Table 1) on sett location, we used a Non-Linear Principal Component Analysis (NLPCA) to reduce the number of variables in each of those categories for the modelling procedure. This approach is equivalent to a standard PCA, but it allowed us to deal with nominal and ordinal variables, often establishing nonlinear relationships (Linting et al., 2007). We included in the model procedure only those principal components that incorporated the majority of data variability associated with the tested variables. NLPCA was carried out using the "homals" R package (de Leeuw and Mair, 2009). Prior to the model building procedure, we tested for correlation significance between all candidate variables not included in NLPCA by assessing Spearman rank correlations (Siegel, 1957). When two variables were significantly correlated, we excluded that which was least correlated with the dependent variable.

To identify the variables influential on badger sett location in our study area, we implemented Conditional Autoregressive Models (CAR) (Dormann et al. 2007), as they account for spatial dependence as a function of a neighbour matrix (see the Moran I index). To test each hypothesis, candidate models were built by combining all variables in each group (i.e. food, disturbance and land cover). Model selection was based on an Information Criterion approach (Burnham and Anderson, 2002). Best models were considered as those with a difference in Akaike Information Criterion scores (with a correction for small sample sizes; AICc) <2 compared to the lowest AICc score (ΔAICc<2; Burnham and Anderson, 2002). As several models fulfilled that criterion, we adopted a model averaging procedure to compute the average of the parameter coefficients included in the best models.

Best model performance was assessed by the Area Under the Curve (AUC) derived from the Receiver Operating Characteristic (ROC) curve (Hanley and McNeil, 1982). Modelling procedures were performed in R (R Core Team, 2014) using the packages "MuMin" (Barton, 2014), "spdep" (Bivand and Piras, 2015) and "pROC" (Robin et al., 2011).

Sett use

We tested for spatial autocorrelation amongst the setts selected for usage analysis, reduced the number of land cover units using a Non-Linear Principal Component Analysis (NLPCA), and tested for variable multicollinearity using Spearman's rank correlations as detailed above. Due to the large number of climate variables considered, some of which were correlated, we decided to take a dimensionality reduction approach using Principal Component Analysis (PCA; Zuur et al., 2007). The PCA was implemented using R software (R Core Team, 2014).

As we had several measurements of sett use for each sett (i.e. data was nested by setts), we implemented a Generalized Linear Mixed Model (GLMM) approach (Zuur et al., 2009) using sett ID as a random factor. We tested all the possible combinations of the considered variables (including the main NLPCA and PCA principal components), using R package "lme4" (Bates et al., 2014; R Core Team, 2014). Model building, selection and performance evaluations were done using the procedures described for sett location.

Results

In total, we detected 18 in-use badger setts (representing a sett density of 0.6 setts/km^2), distributed extensively throughout the study area, although presenting a fairly dense clustering pattern (NNI=0.693; z-score=-2.560, p<0.05). For the analysis of sett location determinants (logistic regression approach), we compared these with 29 non-sett random points in a proportion of 38% vs 62%, respectively.

3.1 Sett location

Spatial autocorrelation was not significant for setts and random points (Moran I=0.040, p=0.07). The NLPCA analysis of land cover and disturbance showed that the first two or three principal components explained the majority of the variance (Land cover D1_LC+D2_LC=0.72; Disturbance D1+D2+D3=0.78; see Table 3 for variable loadings per component). Thus, these five principal components were included as variables in the subsequent analysis. Several variables were correlated ($r_{dist\_c\_wat/Dist\_food}$=0.52, p< 0.001; $r_{dist\_corn/Dist\_wat}$=0.35, p< 0.016), and we excluded those two (Dist_c_wat and Dist_corn) less correlated with the dependent variable.

The best CAR models included two variables associated with food and water availability, and their confidence intervals did not include 0 so we could be sure of the direction of influence (Table 4). These models showed that sett locations were constrained by food-associated variables (Dist_wat and Dist_food). While Dist_food was negatively associated with sett location, the opposite was true for Dist_wat (Table 4), i.e. setts were more often located in areas closer to game species feeding locations and farther away from water sources. These results are supported by good model performance (AUC=0.809).

3.2 Sett use

Only 17 out of the 18 setts were considered in the usage analysis as one had not been surveyed over the entire study period. These seventeen setts were clustered (Moran I=-0.084, p<0.001). The NLPCA and PCA used for the land cover and climatic data showed that the first two components incorporated 71% and 84% of the variable variance, respectively (see Table 5 for variable loadings per component). Due to the significant correlations between several variables ($r_{dist\_wat/Dist\_house}$=0.66, p< 0.001; $r_{Hunt\_seas/C1}$=-0.54, p< 0.001; $r_{Cub/C2}$=0.55, p< 0.001), we excluded three variables from the analysis (Dist_house, Hunt_seas and Cub).

The best models were those associated with disturbance, which included three variables (Table 6). For two of these variables, the confidence intervals did not include 0 so we could infer their importance: Past_P and Unders_cut_p were negatively related to sett use (Table 6), i.e. setts were more often used in periods with low cattle presence and without understory cutting activities. Model performance was AUC=0.61.

Discussion

Our results show that, in the Mediterranean agro-forestry and cattle-raising system we studied, areas closer to game species feeding structures and farther away from water sources have a higher probability of supporting a European badger sett. These findings corroborate our hypothesis that food/water-associated variables constrain sett location. However sett use throughout the year is mainly limited by disturbance, with higher cattle presence and understory-cutting activities hindering sett use. Thus, our analyses indicate that sett location and use are determined by a combination of multi-faceted factors.

These results suggest that although badgers inhabiting western Mediterranean areas have been co-habiting with human activities for centuries, some populations still avoid using areas where cattle breeding and clear-cutting activities (especially understory removal) occur. Setts are essential to this predator's social system, acting as the centre of a clan's territory and activity (Kruuk, 1989). There, adults interact socially (e.g. grooming, mating; Neal and Cheeseman, 1996) and cubs, after first emergence, spend much of their time playing with each other even during the day (Kruuk, 1989).

These behaviours may prompt badgers to use setts located in less disturbed areas, diminishing the risk of encounters with cattle and humans.

The higher probability of setts being located in areas closer to game species feeding structures seems to indicate that, although sensitive to cattle presence and understory cutting at the sett, badgers seem to tolerate game management activities (i.e. food provision to game species in specific structures) in its vicinity. The increased food availability due to such structures (which are regularly filled with wheat) may counter-balance the disturbance effect associated with their maintenance. The importance of food availability for badgers has already been discussed for other Mediterranean badger populations. In southern Portugal, Rosalino et al. (2005) showed that badger setts were not encountered in areas where food patches were absent. Moreover, in Ireland, setts were more abundant in, or near to, high-quality pastures, enhancing the ability of badgers to quickly locate their favoured food, i.e. earthworms (Hammond et al., 2001). Thus, our studied badger population may benefit from the increased food availability near their setts (for example, through shorter feeding journeys).

Our study population sited setts far from water bodies (e.g. streams, usually with riparian galleries). Evidence from other riparian areas is inconsistent with this result, as such landscape patches are typically highly productive and rich feeding grounds for many carnivores (Virgós, 2011). We suggest that other effects may be driving/overriding the perceived benefit of being closer to resource-rich riparian patches and water sources, e.g. to avoid flooding. In the western Mediterranean region, rain is highly concentrated in the autumn/winter season and this increased precipitation can cause a mass influx of water into the ecosystem, occasionally flooding areas in the vicinity of stream beds. Reluctance to locate setts near stream beds to avoid flooding has also been detected for Italian badger populations (Remonti et al., 2006). In that same study, the authors described that during a period of heavy rain the majority of the monitored setts located in the lowlands were flooded.

In conclusion, our study shows that while food availability and flooding risk may constrain sett location, agro-forestry and cattle-raising activities hamper the use of setts by badgers in our study area. These results also indicate that, although European badgers have been co-habiting with humans and their activities for centuries, these factors may still influence how some badger populations use their setts throughout the year. Populations may respond differently to activity types and intensities and, as we have endeavoured to do here, there is a need to identify local trade-offs contributing to biodiversity persistence and sustainable human exploitation of the landscape.

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Figure 1 – Study area location and land cover composition.

Table 1 – Variables used to characterize badger setts and random points according to four main categories (land cover, soil, disturbance and food resources), using two spatial scales (sett/random point or circular area with 350m radius centred at the setts/random points) (NLPCA – Non-Linear Principal Component Analysis).

	Variable	Description	Type
	Sett	Sett/random point	Binary
Land cover (NLPCA)	N_habitat	Number of different habitats within the 350m radius area	Continuous
Land cover (NLPCA)	Main_habit	Habitat with the highest percentage of cover within the 350m radius area	Categorical*
Land cover (NLPCA)	Habitat	Habitat where the setts or random points are located	Categorical*
Land cover (NLPCA)	Veg_herb	Percentage of soil covered by herbaceous plants at the sett or random point	Ordinal*
Land cover (NLPCA)	Veg_shr	Percentage of soil covered by shrubs at the sett or random point	Ordinal*
Land cover (NLPCA)	Veg_arb	Percentage of soil covered by tree canopy at the sett or random point	Ordinal*
	Soil	Type of soil at setts or random point locations	Categorical*
	Dist_house	Distance to the nearest inhabited house	Continuous
Disturbance (NLPCA)	Catt_p	Total number of cattle present in the patch where the sett or random point is located	Ordinal*
Disturbance (NLPCA)	Catt_CA	Mean of the total number of cattle present in the patches overlapping the 350m radius area	Ordinal*
Disturbance (NLPCA)	Tree_harv_p	Evidence of tree harvesting at the sett or random point sites	Binary
Disturbance (NLPCA)	Tree_harv_CA	Evidence of tree harvesting within the 350m radius area	Binary
Disturbance (NLPCA)	Unders_cut_p	Evidence of understory clear-cutting at the sett or random point sites	Binary
Disturbance (NLPCA)	Unders_cut_CA	Evidence of understory clear-cutting within the 350m radius area	Binary
Food-related	Dist_c_wat	Distance to the nearest cattle water supply reservoir (m)	Continuous
Food-related	Dist_food	Distance to the nearest game species feeding locations (m)	Continuous
Food-related	Dist_corn	Distance to the nearest corn production fields (m)	Continuous
Food-related	Dist_wat	Distance to the nearest natural water source (streams) (m)	Continuous

* See Supplementary Material 1 for description of the categories used for each variable.

Table 2 – Variables tested for their influence on European badger sett use probability.

	Variable	Description	Type
	Use	Sett use	Binomial
	Sett	Sett identification	Categorical*
	Cub	Cub emergence season	Binary
	Dist_c_wat	Distance to the nearest water source (natural or cattle reservoir) (m)	Continuous
Land cover (NLPCA)	N_habitat	Number of different habitats within the 350m radius area	Continuous
	Main_habit	Habitat with the highest percentage of cover within the 350m radius area	Categorical*
	Habitat	Habitat where the setts or random points are located	Categorical*
	Veg_herb	Percentage of soil covered by herbaceous plants at the sett or random point	Ordinal*
	Veg_shr	Percentage of soil covered by shrubs at the sett or random point	Ordinal*
	Veg_arb	Percentage of soil covered by tree canopy at the sett or random point	Ordinal*
Disturbance	Dist_house	Distance to the nearest inhabited house	Continuous
	Past_p	Total number of cattle present in the patch where the sett or random point is located	Ordinal*
	Tree_harv_p	Evidence of tree harvesting at the sett or random point	Binary
	Unders_cut_p	Evidence of understory clear-cutting at the sett or random point	Binary
	Hunt_seas	Hunting season	Binary
Climatic (PCA)	Temp_max	Mean daily maximum temperature per month (°C)	Continuous
	Temp_min	Mean daily minimum temperature per month (°C)	Continuous
	Temp_mean	Mean daily temperature per month (°C)	Continuous
	Hum_max	Mean daily maximum air humidity per month (%)	Continuous
	Hum_min	Mean daily minimum air humidity per month (%)	Continuous
	Wind_max	Mean daily maximum wind speed per month (m/s)	Continuous
	Wind_min	Mean daily minimum wind speed per month (m/s)	Continuous
	Prec	Mean daily precipitation per month (mm)	Continuous
	Sol_rad	Mean daily solar radiation per month ($KJ/m^2/day$)	Continuous

* See Supplementary Material 2 for description of the categories used for each variable.

Table 3 – Non-Linear Principal Component Analysis (NLPCA) variable loadings of the Principal Components used in the GLM and CAR models (see descriptions of variables in Table 1).

Land cover			
Original Variable		Loadings	
		D1_LC	D2_LC
N_habitat		0.117	-0.019
Main_habit	Agr	-0.151	0.014
	Shr	0.008	0.043
	COW	0.010	-0.091
	COWU	0.032	-0.005
	Pf	0.009	0.098
	Mf	0.040	0.018
	P	-0.081	0.015
Habitat	Agr	-0.151	0.014
	Shr	0.019	0.041
	COW	0.010	-0.091
	COWU	0.034	-0.012
	Pf	0.005	0.118
	Mf	0.032	0.025
	P	-0.171	-0.013
Veg_herb		0.062	-0.283
Veg_shr		0.229	0.232
Veg_arb		0.267	0.175

Disturbance			
Original Variable	Loadings		
	D1	D2	D3
Catt_p	0.003	0.362	0.104
Catt_CA	-0.109	-0.247	0.091
Tree_harv_p	0.183	-0.123	0.100
Tree_harv_CA	0.256	<-0.001	0.018
Unders_cut_p	0.174	-0.047	0.0119
Unders_cut_CA	0.290	0.009	0.021

Agr - Agricultural fields; Shr – Shrublands; COW - Cork oak woodlands without understory; COWU - Cork oak woodlands with understory; Pf - Pine forests; Mf - Mixed forests; P – Pasture

Table 4 – Characteristics of CAR best models for various hypotheses ordered by increasing AICc values and best model (shaded) averaging parameters (LogLik – Log-likelihood of the linear model; AICc – Akaike Information Criteria with a correction for small sample sizes; w – Akaike weight; β – variable coefficient; SE – Standard Error; P – p-value; CI – Confidence Interval). Bold text – variables with CI (90%) not including 0. See Table 1 for a definition of acronyms. The null model is common to all hypotheses.

	Model	df	LogLik	AICc	ΔAICc	w
	Full model	12	-3.601	129.4		
	Null Model	3	-31.761	70.1		
Land cover	D1_LC	4	-29.021	67.0	0.00	0.546
	D1_LC+D2_LC	5	-28.434	68.3	1.34	0.280
Disturbance	D3	4	-30.552	70.1	0.00	0.148*
	D1	4	-30.990	70.9	0.88	0.096
	D1+D3	5	-29.743	70.9	0.89	0.095
	Dist_house	4	-31.142	71.2	1.18	0.082
	D2	4	-31.404	71.8	1.70	0.063
	D2+D3	5	-30.172	71.8	1.75	0.062
	D1+ Dist_house	5	-30.206	71.9	1.82	0.060
	D3+ Dist_house	5	-30.260	72.0	1.93	0.057
Food and water	Dist_food + Dist_wat	5	-26.455	64.4	0.00	0.497
	Dist_wat	4	-27.927	64.8	0.43	0.400

Model averaging coefficients

	β	SE	P	Upper CI (90%)	Upper CI (90%)
Intercept	0.210	0.146	0.149	-0.029	0.450
Dist_food	**-0.001**	**<0.001**	**0.081**	**-0.004**	**<-0.001**
Dist_wat	**0.004**	**0.001**	**0.007**	**0.002**	**0.008**

* The null model was also included under the Disturbance hypothesis (with a w=0.148 and a ΔAICc =0.03).

Table 5 – Non-Linear Principal Component Analysis (NLPCA) and Principal Component Analysis (PCA) variable loadings of the Principal Components used in the GLMM procedure (see descriptions of variables in Table 2).

Land cover (NLPCA)

Variable		Loadings	
		D1_LC	D2_LC
N_habitat		0.077	0.022
Main_habit	Agr	-0.025	-0.044
	COW	-0.067	0.012
	COWU	0.015	-0.008
	Pf	0.011	0.035
	Mf	0.024	-0.012
Habitat	Agr	-0.025	-0.044
	COW	-0.067	0.012
	COWU	0.015	-0.008
	Shr	0.016	0.009
	Pf	0.006	0.061
	Mf	0.024	-0.012
Veg_herb		-0.340	-0.093
Veg_shr		0.346	-0.015
Veg_arb		0.029	0.316

Climatic (PCA)

Variable	Loadings	
	C1	C2
Temp_max	0.289	-0.396
Temp_min	0.431	
Temp_mean	0.101	-0.489
Hum_max	-0.413	-0.212
Hum_min	0.154	0.498
Wind_max	0.395	0.256
Wind_min	0.398	0.252
Prec	-0.187	0.141

Agr - Agricultural fields; Shr – Shrublands; COW - Cork oak woodlands without understory; COWU - Cork oak woodlands with understory; Pf - Pine forests; Mf - Mixed forests

Table 6 – GLMM model characteristics (ordered by increasing AICc values) for each tested hypothesis and best model (shaded) averaging parameters (LogLik – Log-likelihood of the linear model; AICc – Akaike Information Criteria with a correction for small sample sizes; w – Akaike weight; β – variables coefficient; SE – Standard Error; P – p-value; CI – Confidence Interval) Bold text – variables with CI (90%) not including 0. See Table 2 for a definition of acronyms. The null model is common to all hypotheses.

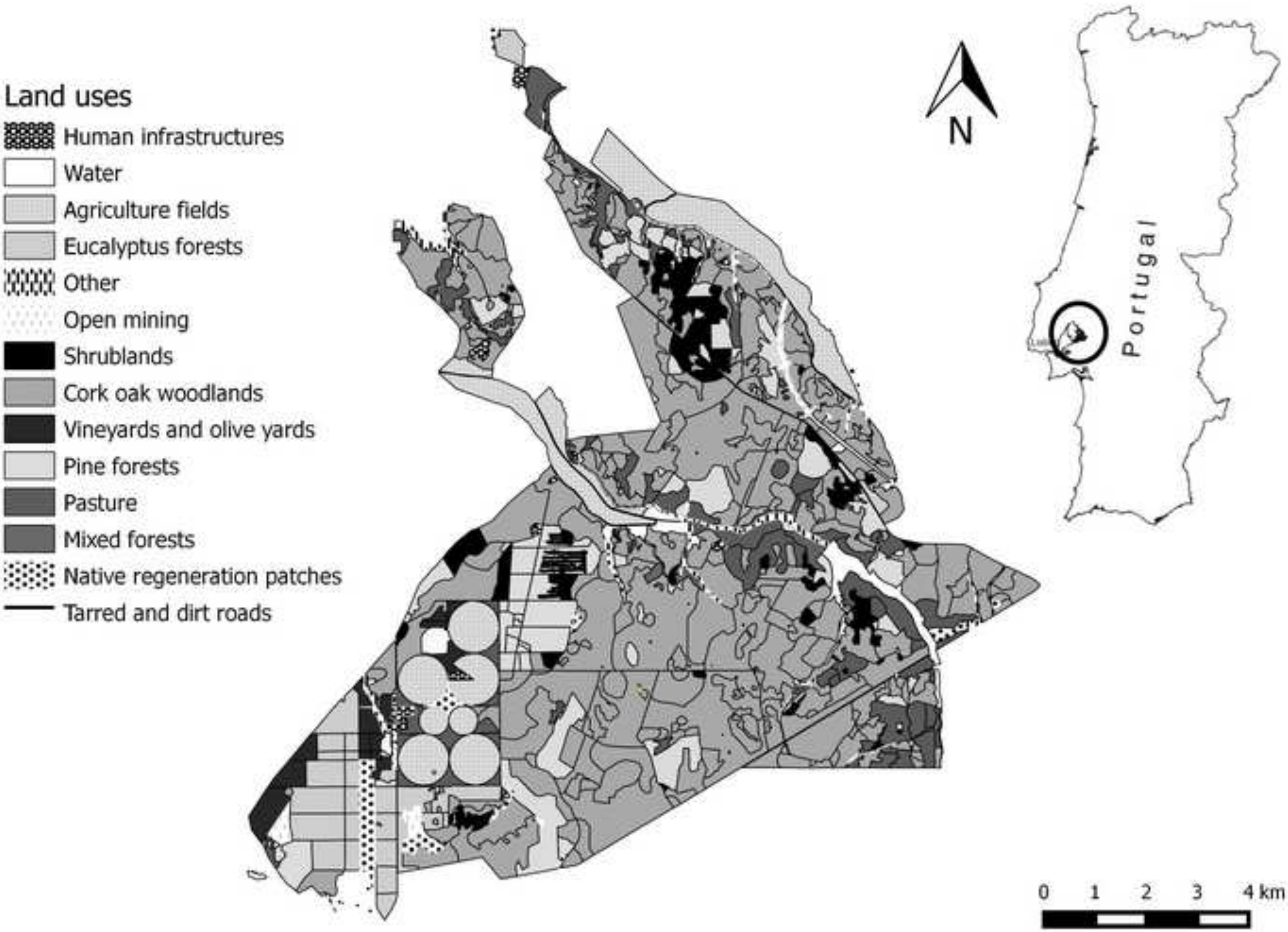
	Model	df	LogLik	AICc	ΔAICc	w
Full model			-70.108	161.4		
Null Model			-84.578	173.2		
Land cover	D1_LC	3	-83.570	173.3	0.04*	0.365
Disturbance	Unders_cut_p + Past_p	4	-73.534	155.3	0.00	0.482
	Past_p	5	-75.292	156.7	1.44	0.235
	Unders_cut_p + Past_p + Tree_harv_p	3	-73.414	157.1	1.86	0.190
Climatic	C1	3	-82.539	171.2	0.00	0.536
Water	Dist_c_wat	3	-84.571	175.3	**	**

Model averaging coefficients

	β	SE	P	Upper CI (90%)	Upper CI (90%)
Intercept	0.652	1.336	0.628	-1.555	2.859
Unders_cut_p (1)	**-2.187**	**2.009**	**0.278**	**-5.902**	**-0.002**
Past_p	**-1.684**	**0.507**	**0.001**	**-2.522**	**-0.847**
Tree_harv_p (1)	-0.118	0.585	0.841	-2.506	1.383

*Compared to the null model that was the best model of the land cover hypothesis; ** only one variable was used, so no ΔAICc or w values were calculated

Figure 1


Land uses
Human infrastructures
Water
Agriculture fields
Eucalyptus forests
Other
Open mining
Shrublands
Cork oak woodlands
Vineyards and olive yards
Pine forests
Pasture
Mixed forests
Native regeneration patches
Tarred and dirt roads
N
Portugal
0 1 2 3 4 km